

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408597
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Gamble; Jeffrey

Plant protector with slidingly adjustable and telescoping handle

Abstract

The plant protector may comprise a plant shield and a handle. The plant protector may be placed adjacent to a plant in preparation for using a lawn maintenance tool. As a non-limiting example, the lawn maintenance tool may be a string trimmer. The plant shield may protect the plant from being damaged by the operation of the lawn maintenance tool. The handle may be adapted to be grasped by a user while placing the plant protector. The length of the handle may telescopically adjust to be longer or shorter.

Inventors:	Gamble; Jeffrey (Chelsea, MI)
Applicant:	Gamble; Jeffrey (Chelsea, MI)
Family ID:	96950318
Appl. No.:	18/636175
Filed:	April 15, 2024

Publication Classification

Int. Cl.: A01G13/27 (20250101); A01G13/29 (20250101)

U.S. Cl.:

CPC A01G13/27 (20250101); A01G13/29 (20250101);

Field of Classification Search

CPC: A01G (13/27)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1955950	12/1933	Copeman	118/DIG.6	A01G 13/10
4845889	12/1988	Taylor	N/A	N/A
4995191	12/1990	Davis	47/32.7	A01G 29/00
5048229	12/1990	Campbell	47/32.4	A01G 13/27
5385005	12/1994	Ash	N/A	N/A
5594990	12/1996	Brant	172/14	B25G 1/04
11457572	12/2021	Battoe	N/A	N/A
11464175	12/2021	Clarke	N/A	A01G 13/27
2009/0126298	12/2008	Salcedo	N/A	N/A
2016/0095284	12/2015	Britt	N/A	N/A
2016/0270305	12/2015	Shaffer et al.	N/A	N/A
2018/0235157	12/2017	Leslie	N/A	N/A
2019/0246867	12/2018	Ziamandanis	N/A	B25G 1/04
2019/0261579	12/2018	Brown	N/A	N/A
2020/0337230	12/2019	Bian	N/A	A01D 34/475
2021/0228390	12/2020	Shepherd	N/A	A61F 4/00
2022/0061233	12/2021	Peace	N/A	N/A
2022/0251870	12/2021	DeWitt	N/A	N/A

Primary Examiner: Wong; Jessica B

Attorney, Agent or Firm: Cramer Patent & Design, PLLC.

Background/Summary

RELATED APPLICATIONS

(1) None.

FIELD OF THE DEVICE

(2) The presently disclosed subject matter is directed to the field of gardening tools and, more specifically, to devices designed to protect plants during the operation of lawn maintenance tools. The device pertains to a plant protector that offers a safeguard against damage from such tools, enabling users to maintain the area around plants without harming them. This device is particularly useful in the context of residential and commercial landscaping where the proximity of valuable or delicate plants to areas requiring maintenance poses a challenge.

BACKGROUND OF THE DEVICE

(3) In the maintenance of lawns and gardens, it is often necessary to use tools such as string trimmers, edgers, and mowers. While these tools are effective for their intended purpose, they can also pose a risk to nearby plants, causing physical damage that can impair growth or even result in the death of the plant. Traditional methods of protecting plants, such as manually holding a barrier or avoiding the area altogether, are either ineffective or result in less than satisfactory maintenance of the lawn. There exists a need for a device that can protect plants from damage caused by lawn maintenance tools, which is easy to deploy and adjust by a single user, and which can adapt to various sizes and shapes of plants. The plant protector as described herein seeks to address these needs by providing a robust, adjustable, and easy-to-use device that shields plants from potential harm during lawn and garden maintenance activities.

SUMMARY OF THE DEVICE

(4) Embodiments of the present disclosure may include a plant protector, including a plant shield configured to form a barrier between a plant and a lawn maintenance tool. In some embodiments, the plant shield may include a semi-cylindrical shell defining a concave inner surface facing the plant and a convex outer surface exposed to the lawn maintenance tool. Embodiments may also include a handle coupled to the plant shield. In some embodiments, the handle may be telescopic, enabling adjustment of the handle's length.

(5) In some embodiments, the plant shield may be tapered, including a top portion with a smaller radius than a bottom portion of the plant shield. In some embodiments, the plant shield may include two or more horizontal slots parallel to each other. In some embodiments, the handle may be repositionable on the plant shield by sliding the handle along the horizontal slots.

(6) In some embodiments, the handle may include a plurality of telescoping sections enabling the handle to collapse for storage and extend for use. In some embodiments, the plant protector may include a handle adapter. In some embodiments, the handle may be coupled to the plant shield via the handle adapter. In some embodiments, the handle adapter may be coupled to the plant shield using mounting fasteners.

(7) In some embodiments, the mounting fasteners include at least one of screws, nuts, bolts, washers, rivets, hooks, collars, nipples, cams, standoffs, knobs, and brackets. In some embodiments, the handle may include a strap enabling the plant protector to be secured to a fixed object. In some embodiments, the plant shield may be made of molded plastic.

(8) In some embodiments, the plant shield may be made of high-density polyethylene (HDPE). In some embodiments, the plant protector, configured for use in conjunction with additional plant protectors of the same configuration. In some embodiments, each plant protector may be placed adjacent to a different plant to allow operation of a lawn maintenance tool between the plants without damaging them.

(9) In one specific embodiment, the plant protector consists of a plant shield designed to safeguard plants during lawn maintenance activities. The plant shield features a semi-cylindrical shell with a concave inner surface tailored to face the plant, and a convex outer surface to interact with the environment and deflect potential damage from lawn maintenance tools. Incorporated within the plant shield are horizontal slots, which facilitate the adjustable attachment of a handle, allowing for repositioning based on user preference or specific task requirements. The handle, essential for maneuvering the plant protector, includes telescoping sections to adjust the handle's length for ease of storage or to extend its reach during use. This adjustability enhances the ergonomic handling and versatility of the device. A handle adapter serves as the intermediary for securing the handle to the plant shield, employing mounting fasteners to ensure a robust and durable connection. Additionally, a strap attached to the handle provides the option to secure the plant protector to a stationary object, offering stability and hands-free convenience when necessary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The advantages and features of the present device will become better understood with reference to the following more detailed description and claims taken in conjunction with the accompanying drawings, in which like elements are identified with like symbols, and in which:

(2) FIG. 1 is a front isometric view of a plant protector **100**, according to an embodiment of the present device;

(3) FIG. 2 is a rear isometric view of a plant protector **100**, according to an embodiment of the present device;

(4) FIG. 3 is a front view of a plant protector **100**, according to an embodiment of the present device;

- (5) FIG. 4 is a rear view of a plant protector **100**, according to an embodiment of the present device;
- (6) FIG. 5 is a left side view of a plant protector **100**, according to an embodiment of the present device;
- (7) FIG. 6 is a right side view of a plant protector **100**, according to an embodiment of the present device;
- (8) FIG. 7 is a top view of a plant protector **100**, according to an embodiment of the present device;
- and,
- (9) FIG. 8 is a bottom view of a plant protector **100**, according to an embodiment of the present device.

DESCRIPTIVE KEY

(10) **100** plant protector **200** plant shield **206** horizontal slot **210** concave inner surface **212** convex outer surface **220** handle **222** handle adapter **224** mounting fastener **226** telescoping section **240** strap

DESCRIPTION OF THE DEVICE

- (11) The present device is directed to a plant protector (herein described as the “device”) **100**. The device **100** may comprise a plant shield **200** and a handle **220**. The device **100** may be placed adjacent to a plant in preparation for using a lawn maintenance tool. As a non-limiting example, the lawn maintenance tool may be a string trimmer. The plant shield **200** may protect the plant from being damaged by the operation of the lawn maintenance tool. The handle **220** may be adapted to be grasped by a user while placing the device **100**. The length of the handle **220** may telescopically adjust to be longer or shorter.
- (12) The plant shield **200** may form a barrier between the plant and the lawn maintenance tool. The plant shield **200** may be the shape of semi-cylindrical shell defining a concave inner surface **210** and a convex outer surface **212**. The concave inner surface **210** may face the plant. The convex outer surface **212** may be exposed to the lawn maintenance tool.
- (13) In some embodiments, the plant shield **200** may be tapered such that the top of the plant shield **200** may comprise a smaller radius than the bottom of the plant shield **200**.
- (14) In some embodiments, the plant shield **200** may comprise two (2) or more horizontal slots **206**. The horizontal slots **206** may be parallel to each other. The handle **220** may couple to the plant shield **200** via the horizontal slots **206**. The handle **220** may be repositionable on the plant shield **200** by sliding the handle **220** along the horizontal slots **206**.
- (15) The handle **220** may be coupled to the plant shield **200** and may extend upwards. The handle **220** may be adapted for the user to grasp while placing the plant shield **200**.
- (16) The handle **220** may comprise a plurality of telescoping sections **226** such that the handle **220** may telescopically vary in length. As non-limiting examples, the handle **220** may telescopically collapse for storage and may telescopically extend for use.
- (17) In some embodiments, the handle **220** may couple to the plant shield **200** via a handle adapter **222**. Specifically, the bottom of the handle **220** may couple to the top of the handle adapter **222** and the handle adapter **222** may couple to the plant shield **200** using mounting fasteners **224**. As non-limiting examples, the mounting fasteners **224** may comprise screws, nuts, bolts, washers, rivets, hooks, collars, nipples, cams, standoffs, knobs, brackets, or any combination thereof.
- (18) In some embodiments, the handle **220** may comprise a strap **240** such that the device **100** may be secured to a tree, a fence, or other fixed object.
- (19) The plant shield **200** may be molded plastic. As a non-limiting example, the plant shield **200** may be made of high density polyethylene (HDPE).
- (20) Multiple shields **200** may be used at the same time to protect multiple plants that may be in close proximity to each other. As a non-limiting example, a first shield **200** may be placed adjacent to a first plant and a second shield **200** may be placed adjacent to a second plant so that the lawn maintenance tool may be operated between the first plant and the second plant without damaging

either plant.

(21) In use, the device **100** may be deployed by placing the plant shield **200** adjacent to a plant with the concave inner surface **210** facing towards the plant and the convex outer surface **212** facing away from the plant. The handle **220** may be extended and used to place the plant shield **200**. A lawn maintenance tool such as a string trimmer may be operated near the plant without damaging the plant. The handle **220** may be collapsed while storing the device **100**.

(22) In one specific embodiment of the plant protector (**100**), the plant protector consists of a plant shield (**200**) designed to safeguard plants during lawn maintenance activities. The plant shield (**200**) features a semi-cylindrical shell with a concave inner surface (**210**) tailored to face the plant, and a convex outer surface (**212**) to interact with the environment and deflect potential damage from lawn maintenance tools. Incorporated within the plant shield (**200**) are horizontal slots (**206**), which facilitate the adjustable attachment of a handle (**220**), allowing for repositioning based on user preference or specific task requirements. The handle (**220**), essential for maneuvering the plant protector (**100**), includes telescoping sections (**226**) to adjust the handle's length for ease of storage or to extend its reach during use. This adjustability enhances the ergonomic handling and versatility of the device. A handle adapter (**222**) serves as the intermediary for securing the handle (**220**) to the plant shield (**200**), employing mounting fasteners (**224**) to ensure a robust and durable connection. Additionally, a strap (**240**) attached to the handle (**220**) provides the option to secure the plant protector (**100**) to a stationary object, offering stability and hands-free convenience when necessary.

(23) The exact specifications, materials used, and method of use of the device **100** may vary upon manufacturing. The foregoing descriptions of specific embodiments of the present device have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the device to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teaching. The embodiments were chosen and described in order to best explain the principles of the device and its practical application, to thereby enable others skilled in the art to best utilize the device and various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A plant protector, comprising: a plant shield configured to form a barrier between a plant and a lawn maintenance tool, wherein the plant shield comprises a semi-cylindrical shell defining a concave inner surface facing the plant and a convex outer surface exposed to the lawn maintenance tool; and, a handle coupled to the plant shield, wherein the handle is telescopic, enabling adjustment of the handle's length; and, wherein the plant shield comprises two or more horizontal slots parallel to each other, and wherein the handle is repositionable on the plant shield by sliding the handle along the horizontal slots.
2. The plant protector of claim 1, wherein the plant shield is tapered, comprising a top portion with a smaller radius than a bottom portion of the plant shield.
3. The plant protector of claim 1, wherein the handle comprises a plurality of telescoping sections enabling the handle to collapse for storage and extend for use.
4. The plant protector of claim 1, further comprising a handle adapter, wherein the handle is coupled to the plant shield via the handle adapter, and wherein the handle adapter is coupled to the plant shield using mounting fasteners.
5. The plant protector of claim 4, wherein the mounting fasteners include at least one of screws, nuts, bolts, washers, rivets, hooks, collars, nipples, cams, standoffs, knobs, and brackets.
6. The plant protector of claim 1, wherein the plant shield is made of molded plastic.
7. The plant protector of claim 6, wherein the plant shield is made of high density polyethylene (HDPE).

8. The plant protector of claim 1, configured for use in conjunction with additional plant protectors of the same configuration, wherein each plant protector is placed adjacent to a different plant to allow operation of a lawn maintenance tool between the plants without damaging them.
